

TAKUR SAI KARTHIK CHALAMALASETTY

3938 Radcliff Drive, Canton, Michigan 48188

📞 734-776-7941

✉ chtakur1@gmail.com

🌐 linkedin.com/in/karthik-chalamalasetty

Summary

Recent graduate from Lawrence Technological University with a Master's in Computer Science (Dec 2025). Skilled in Python, SQL, Machine Learning, and AI-based automation workflows. Currently on F-1 OPT and seeking roles in Data Science, AI, Machine Learning, or Automation; will require H-1B sponsorship in the future.

Education

Lawrence Technological University

Master of Science in Computer Science

Jan 2024 – Dec 2025

Southfield, MI

SRM University AP

Bachelor of Technology in Computer Science and Engineering

June 2019 – May 2023

Amaravati, India

Experience

SRM University AP

Web Designing Intern

May 2022 – May 2023

Amaravati, India

- Built a website for the International Relations department, increasing user engagement by 20% and improving access to key resources.
- Applied data analysis techniques to improve user interaction insights and optimize content delivery.
- Used Agile methodology to streamline development, completing 10+ sprints with a 98% on-time delivery rate.
- Collaborated with cross-functional teams to create responsive web features, improving accessibility for 500+ international students.

Tevatron Technologies Pvt. Ltd. (Hexnbit)

Artificial Intelligence & Machine Learning Intern

Jun 2021 – Jul 2021

Remote, India

- Completed a 45-day internship focused on Artificial Intelligence and Machine Learning concepts and applications.
- Developed a project on Wine Quality Detection using AI/ML models, applying machine learning techniques for classification and data analysis.
- Gained hands-on experience with AI/ML workflows including data preprocessing, model training, and evaluation.
- Recognized for strong analytical skills and consistent performance throughout the internship.

Technical Skills

Languages: Python, Java, SQL

Developer Tools: Git, GitHub, Jupyter, Visual Studio, Eclipse, Postman, Google Colab

Data Science & Machine Learning: NumPy, Pandas, scikit-learn, OpenCV, Statistical Analysis, Data Visualization, TensorFlow, Keras, PyTorch, YOLOv8, Matplotlib, MLflow, NLP, Transformer Models, LLM Evaluation

AI & Automation: Prompt Engineering, LLM APIs (OpenAI, Claude), REST APIs, Data Automation Workflows

Cloud & Tools: AWS (S3, SageMaker, EC2)

Coursework: Artificial Intelligence, Machine Learning, Big Data, Advanced Data Structures, Advanced Data Science, Software Engineering, Algorithm Design & Analysis, Database Systems, Web Server Programming, Data Mining, Theory of Computation

Projects

AI Automation System | LLM, Python, APIs

- Built an AI-powered automation system to process unstructured data using LLM APIs and Python.
- Automated data summarization and extraction using Python and APIs.
- Designed prompt-based workflows to convert raw data into structured outputs.
- Automated report generation and structured output creation for downstream analysis.
- Integrated data pipelines using Python and handled real-world data inputs.
- Reduced manual data processing effort by 60% through automation.

LLM Response Evaluation System | Python, PyTorch, NLP, Transformers

- Built an LLM-based system to evaluate and automate analysis of model responses.
- Developed scripts to process multiple prompts and generate structured evaluation reports.
- Detected hallucinations and inconsistencies in model outputs using automated pipelines.
- Improved evaluation efficiency by reducing manual review effort.

Automated Detection and Classification of Underwater Species | *Python, YOLOv8, Deep Learning, OpenCV*

- Built a real-time underwater species detection and segmentation system using YOLOv8 for automated monitoring in marine environments.
- Preprocessed and augmented a multi-class underwater dataset to handle low visibility, class imbalance, and noise, improving robustness.
- Achieved 80.82% Precision, 69.35% Recall, 76.86% mAP@0.5, and 85.48% segmentation accuracy using transfer learning and optimized feature extraction.

Security of Drone-Based Delivery Systems | *Python, Flask, Cryptography, Simulation, Data Analysis*

- Designed a data-driven three-layer security framework for drone communication, delivery channels, and customer verification to reduce last-mile risks.
- Built a simulation using AES-256 encryption, SHA-256 hashing, and OTP authentication; generated security logs and performance metrics across attack scenarios.
- Achieved 100% attack prevention with low overhead (<50 ms encryption latency, 30.6 s average delivery time).

AI-Based Vehicle Fault Detection and Predictive Maintenance System | *Python, Machine Learning, Pandas, scikit-learn*

- Built a predictive maintenance system to analyze automotive sensor data and detect abnormal vehicle behavior.
- Processed sensor data (speed, RPM, engine temperature) to identify early fault patterns using anomaly detection.
- Implemented and evaluated ML models to predict failures, enabling proactive maintenance and improving reliability.
- Engineered a data preprocessing and feature extraction pipeline to improve model accuracy and enable real-time vehicle health monitoring.

Publications

Takur Sai Karthik Chalamalasetty, *Automated Detection and Classification of Underwater Species Using YOLOv8 for Real-time Marine Ecosystem Monitoring*, ResearchGate, 2025.

Available at: researchgate.net/publication/398553650